TTK4190 Guidance, Navigation and Control of Marine Craft, Aircraft and Drones

Assignment 2 part 2

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Problem 1 - Environmental Disturbances

Problem 1a

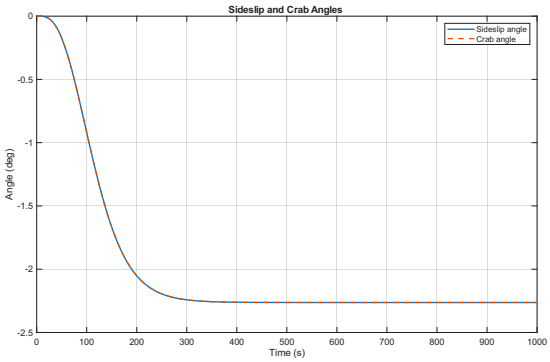
The oceean current is modeled as a 2-D irrotinal current with constant speed $V_c = 1\text{m/s}$ and direction $\beta_{V_C} = 45°$. The current velocity is transformed from the Earth-fixed coordinate system to the ship's body-fixed coordinate system using

$$\begin{aligned} u_c &= V_C \cos(\beta_{V_C} - \psi) \\ v_c &= V_C \sin(\beta_{V_C} - \psi) \end{aligned} \tag{1}$$

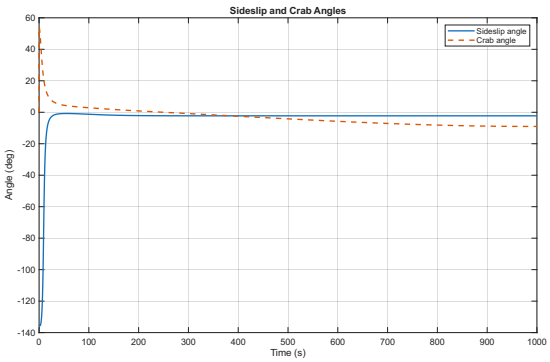
Problem 1b

The sideslip angle is calculated using the ship's velocity relative to the water, while the crab angle uses the velocity relative to the Earth

$$\begin{aligned} \beta &= \text{atan2}(v - v_c, u - u_c) \\ \beta_c &= \text{atan2}(v, u) \end{aligned} \tag{2}$$



(a) Sideslip and crab angles without current



(b) Sideslip and crab angles with current

Without current, the sideslip and crab angles are identical, since the ship's velocity relative to the water is equal to its velocity relative to the Earth. Both angle settle around $-2.25°$. With a current of 1m/s towards the north-east, the angles differ. The sideslip angle settles around $-2.2°$., while the crab angle reaches around $-9°$.

This difference occurs because the ocean current changes the ship's direction of movement relative to the Earth, while the sideslip angle describes its movement relative to the surrounding water. The large initial angles occur because the ship starts at low speed, where small velocity changes can cause large changes in the calculated angles.The large initial angles occur because the ship starts at low speed, where small velocity changes can cause large changes in the calculated angles.

Problem 1c

The wind disturbances affect the ship in sway and yaw. The forces depend on the relative wind speed and direction. The relative wind velocity is calculated by subtracting the ship's velocity from the wind, velocity, expressed in the same coordinate frame.

The sway force and yaw moment are calculated by these equations from Fossen [1]:

$$\begin{aligned} Y_{wind} &= \frac{1}{2}\rho_a V_{rw}^2 C_Y(\gamma_{rw}) A_{L_w} \\ &= \frac{1}{2}\rho_a V_{rw}^2 A_{L_w} c_y \sin(\gamma_{rw}) \\ N_{wind} &= \frac{1}{2}\rho_a V_{rw}^2 C_N(\gamma_{rw}) A_{L_w} L_{oa} \\ &= \frac{1}{2}\rho_a V_{rw}^2 A_{L_w} L_{oa} c_n \sin(\gamma_{rw}) \end{aligned} \qquad \begin{aligned} V_{rw} &= \sqrt{u_{rw}^2 + \nu_{rw}^2} \\ \gamma_{rw} &= -\operatorname{atan2}(\nu_{rw}, u_{rw}) \end{aligned} \tag{3}$$

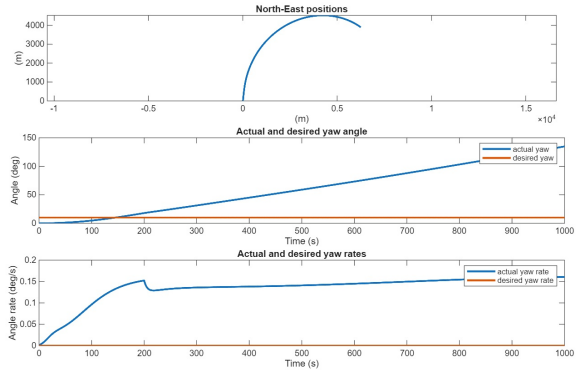


Figure 2: Graphs after adding wind moments after 200s

The wind disturbance is introduced at $t = 200$s, causing a noticeable change in the ship's yaw rate. This shows the wind moment affects the ship's rotational motion.

Problem 2 - Heading Autopilot

Problem 2a

The Coriolis forces are linearized around the operating point $v = r = 0$ and $u = u_d$. Starting with the nonlinear expressions from assignment 2 part 1[2].

$$
\begin{aligned}
\mathbf{C_{RB}}(\nu)\nu &= \begin{bmatrix} 0 & 0 & -m\left(v + rx_g\right) \\ 0 & 0 & mu \\ m\left(v + rx_g\right) & -mu & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} \\
&= \begin{bmatrix} -mx_g r^2 - mvr \\ mur \\ mx_g ur \end{bmatrix} \\
\mathbf{C_A}(\nu)\nu &= \begin{bmatrix} 0 & 0 & Y_{\dot{v}} v + Y_{\dot{r}} r \\ 0 & 0 & -X_{\dot{u}} u \\ -Y_{\dot{v}} v - Y_{\dot{r}} r & X_{\dot{u}} u & 0 \end{bmatrix} \begin{bmatrix} u \\ v \\ r \end{bmatrix} \\
&= \begin{bmatrix} Y_{\dot{v}} vr + Y_{\dot{r}} r^2 \\ -X_{\dot{u}} ur \\ (X_{\dot{u}} - Y_{\dot{v}}) uv - Y_{\dot{r}} ur \end{bmatrix}
\end{aligned} \tag{4}
$$

The nonlinear Coriolis forces are linearized around the operating point

$$
u = u_d \qquad v = r = 0 \tag{5}
$$

by introducing the deviations

$$
u = u_d + \Delta u \qquad v = \Delta v \qquad r = \Delta r \tag{6}
$$

and neglecting second-order terms.

Substituting the deviations into the rigid-body Coriolis forces gives

$$
\begin{aligned}
\mathbf{C_{RB}}(\nu)\nu &= \begin{bmatrix} -mx_g(\Delta r)^2 - m\Delta v \Delta r \\ m(u_d + \Delta u)\Delta r \\ mx_g(u_d + \Delta u)\Delta r \end{bmatrix} \\
&\approx \begin{bmatrix} 0 \\ mu_d r \\ mx_g u_d r \end{bmatrix} = \mathbf{C^*_{RB}}\nu
\end{aligned} \tag{7}
$$

Therefor the linearized rigid-body Coriolis matrix is

$$
\mathbf{C^*_{RB}} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & mu_d \\ 0 & 0 & mx_g u_d \end{bmatrix} \tag{8}
$$

For the added-mass Coriolis forces, we start with

$$
\begin{aligned}
\mathbf{C_A}(\nu)\nu &= \begin{bmatrix} Y_{\dot{v}} \Delta v \Delta r + Y_{\dot{r}} (\Delta r)^2 \\ -X_{\dot{u}}(u_d + \Delta u)\Delta r \\ (X_{\dot{u}} - Y_{\dot{v}})(u_d + \Delta u)\Delta v - Y_{\dot{r}}(u_d + \Delta u)\Delta r \end{bmatrix} \\
&\approx \begin{bmatrix} 0 \\ -X_{\dot{u}} u_d r \\ (X_{\dot{u}} - Y_{\dot{v}}) u_d v - Y_{\dot{r}} u_d r \end{bmatrix} = \mathbf{C^*_A}\nu
\end{aligned} \tag{9}
$$

Therefor, the linearized added-mass Coriolis matrix is

$$
\mathbf{C^*_A} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -X_{\dot{u}} u_d \\ 0 & (X_{\dot{u}} - Y_{\dot{v}}) u_d & -Y_{\dot{r}} u_d \end{bmatrix} \tag{10}
$$

The resulting matrices represent the first-order approximation of the Coriolis forces around the chosen operating point.

Problem 2b

The linear sway-yaw maneuvering model is given by

$$\mathbf{M}\dot{\nu}_\mathbf{r} + \mathbf{N}\nu_\mathbf{r} = \mathbf{b}\delta \tag{11}$$

where

$$\begin{aligned} \mathbf{M} &= \mathbf{M_{RB}} + \mathbf{M_A} \\ \mathbf{N} &= \mathbf{C^*_{RB}} + \mathbf{C^*_A} + \mathbf{D} \\ \mathbf{b} &= 2u_d \begin{bmatrix} -Y_\delta & -N_\delta \end{bmatrix}^\top \end{aligned} \tag{12}$$

Rearranging the equations gives

$$\dot{\nu} = -\mathbf{M^{-1}}\mathbf{N}\nu_\mathbf{r} + \mathbf{M^{-1}}\mathbf{b}\delta \tag{13}$$

Comparing this with the standard state-space representation $\dot{x} = Ax + Bu$, we obtain

$$A = -\mathbf{M^{-1}}\mathbf{N}, \qquad B = \mathbf{M^{-1}}\mathbf{b} \tag{14}$$

Since the desired output is the yaw rate r, we choose

$$C = \begin{bmatrix} 0 & 1 \end{bmatrix} \qquad D_{ss} = 0 \tag{15}$$

The transfer function from rudder angle to yaw rate is therefore

$$G(s) = \frac{r(s)}{\delta(s)} = C(s\mathbf{I} - A)^{-1}B \tag{16}$$

The matrices $\mathbf{M}$ and $\mathbf{N}$ are calculated using the parameters from ship.m. For both matrices, only the sway and yaw components are used (rows and columns 2 and 3). The input vector $\mathbf{b}$ is calculated using the rudder coefficients Y_δ and N_δ from ship.m, with the desired surge velocity $u_d = 7$. We obtain:

$$\begin{aligned} \mathbf{M} &\approx \begin{bmatrix} 2.22673 \cdot 10^7 & -6.408626 \cdot 10^7 \\ -6.408626 \cdot 10^7 & 4.6015 \cdot 10^{10} \end{bmatrix} \\ \mathbf{N} &\approx \begin{bmatrix} 1.1134 \cdot 10^6 & 1.258 \cdot 10^8 \\ 3.01 \cdot 10^7 & 4.1529 \cdot 10^9 \end{bmatrix} \\ \mathbf{b} &\approx \begin{bmatrix} -5.5979 \cdot 10^4 \\ 4.1200 \cdot 10^6 \end{bmatrix} \end{aligned} \tag{17}$$

Using MATLAB's $ss2tf(A, B, C, D_{ss})$ function we obtain the numerical transfer function

$$G(s) = \frac{8.638 \cdot 10^{-5}s + 6.146 \cdot 10^{-6}}{s^2 + 0.1506s + 8.203 \cdot 10^{-4}} \tag{18}$$

Problem 2c

The second order Nomoto form is

$$G(s) = \frac{r(s)}{\delta(s)} = \frac{K(T_3 s + 1)}{(T_1 s + 1)(T_2 s + 1)} \tag{19}$$

We can normalize the transfer function by dividing the numerator and denominator by $8.203 \cdot 10^{-4}$

$$G(s) = \frac{0.10530s + 0.007493}{1219.07s^2 + 183.59s + 1} \tag{20}$$

Comparing this with

$$G(s) = \frac{KT_3 s + K}{T_1 T_2 s^2 + (T_1 + T_2)s + 1} \tag{21}$$

We identify

$$K = 0.007493 \quad T_1 + T_2 = 183.59 \quad KT_3 = 0.10530 \tag{22}$$

$$T_3 = \frac{0.10530}{0.007493} = 14.06 \tag{23}$$

$$T = T_1 + T_2 - T_3 = 183.59 - 14.06 = 169.53$$

The transfer function becomes

$$\frac{r}{\delta}(s) = \frac{0.007493}{169.53s + 1} \tag{24}$$

Problem 2d

The first-order Nomoto model derived in Problem 2c is

$$T\dot{r} + r = K\delta$$

where $T = 169.53\,\mathrm{s}$ and $K = 0.007493\,\mathrm{s}^{-1}$.

A PID heading controller is designed using pole placement with $\omega_n = 0.06\,\mathrm{rad/s}$ and $\zeta = 1$. The control law is

$$\delta_c = K_p e_\psi + K_d e_r + K_i e_{\mathrm{int}}$$

where

$$e_\psi = \psi_d - \psi, \qquad e_r = r_d - r, \qquad \dot{e}_{\mathrm{int}} = e_\psi$$

The closed-loop characteristic polynomial becomes

$$Ts^3 + (1 + KK_d)s^2 + KK_p s + KK_i = 0$$

Choosing the desired characteristic polynomial as

$$(s + \omega_n)(s^2 + 2\zeta\omega_n s + \omega_n^2)$$

and comparing coefficients gives

$$K_p = \frac{T(1 + 2\zeta)\omega_n^2}{K}$$

$$K_d = \frac{T(2\zeta + 1)\omega_n - 1}{K}$$

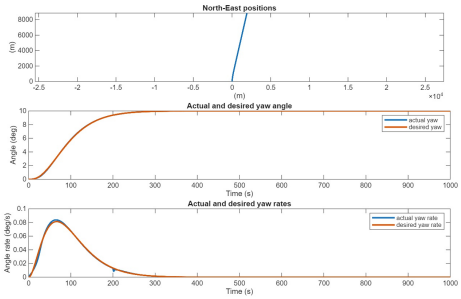
$$K_i = \frac{T\omega_n^3}{K}$$

Using the specified parameters gives

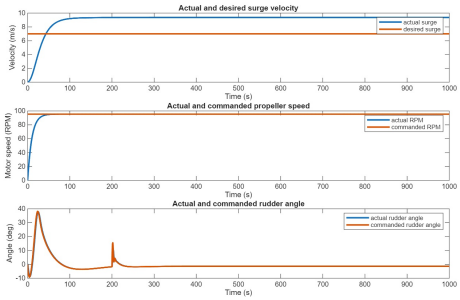
$$K_p \approx 244.4, \quad K_d \approx 3938, \quad K_i \approx 4.887$$

The simulation in Figure 3a and 3b shows that the actual heading closely follows the desired heading and reaches 10°. The yaw rate also follows the reference model and approaches zero as the ship reaches the desired heading.

When the wind disturbance is introduced at $t = 200$ s, a small deviation in the yaw rate occurs, and the rudder angle changes to compensate. The heading remains close to the desired value, showing that the PID controller can compensate for the environmental disturbances.



(a) Actual and desired heading and yaw rate using the PID heading controller.



(b) Actual and commanded rudder angles, showing the wind disturbance at t = 200s.

Problem 2e

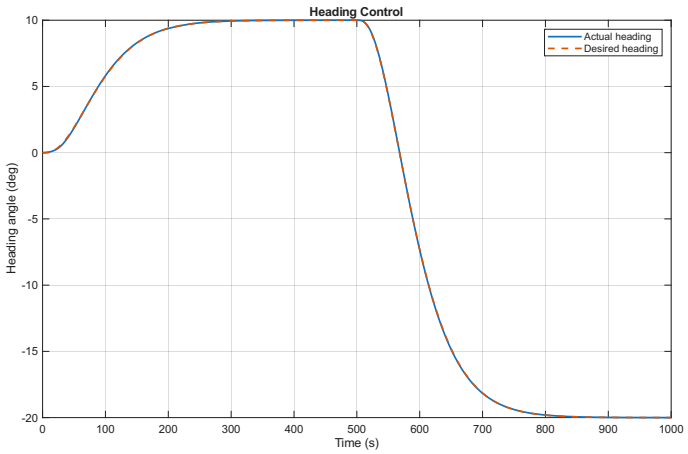


Figure 4: Heading control with setpoint 10° then −20° at t > 500s.

Figure 4 shows the heading response for a 10° setpoint, followed by a change to −20° at $t = 500$ s. The actual heading closely follows the desired heading, with negligible overshoot and steady-state error. This indicates good tracking performance of the PID controller.

From Figure 5 we can se that the rudder angle reaches approximately 38° initially and −27° after the setpoint change, remaining within the saturation limits of ±40°. The controller shows no significant overshoot or delayed recovery, indicating that integrator windup is not a problem in these simulations. Therefore, no anti-windup strategy is implemented.

Problem 3 - Propeller Revolution and Speed Control

Problem 3a

The propeller thrust and torque coefficients was calculated once using the Matlab function `[KT,KQ] = wageningen(J,`

```
Ja = 0;
P_D = 1.5;        %pitch/diameter ratio
AE_A0 = 0.65;     %blade area ratio
Z = 4;            %number of blades
[KT,KQ] = wageningen(Ja, P_D, AE_A0, Z);
```

This resulted in the trust coefficient $KT = 0.6367$ and the torque coefficient $KQ = 0.1390$. Both KT and KQ was set manually for later executions.

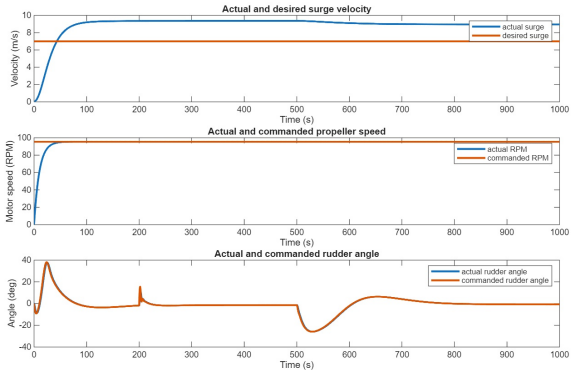


Figure 5: Rudder angle with setpoint 10° then −20° at t > 500s.

Problem 3b

The relationship between motor torque and generated shaft speed is described in Fossen[1]:

$$Q = \rho D^5 K_Q(J_A)|n|n \tag{25}$$

Hence, for a given desired motor shaft speed n_d, the equivalent torque needed to generate that speed is given by:

$$Q_d = \rho D^5 K_Q(J_A)|n_d|n_d \tag{26}$$

Problem 3c

From Figure 1 from the assignment description [2], we observe that

$$Y = \frac{1}{K_m} Q_d(n_d).$$

The transfer function H(s) can be found in Fossen[1] equation 9.35. Using this transfer function, combined with the information given in the task and the expression for Y, it is possible to derive the expression for $\dot{Q}_m$:

$$\begin{aligned}
Q_m &= H(s)Y \\
Q_m &= \frac{K_m}{sT_m + 1} e^{-\tau s} Y \qquad \text{where } \tau = 0 \\
Q_m &= \frac{1}{sTm + 1} Q_d \\
(sTm + 1)Q_m &= Q_d \\
\mathcal{L}^{-1}\{(sTm + 1)Q_m &= Q_d\} \\
\dot{Q}_m &= \frac{1}{T_m}(Q_d - Q_m)
\end{aligned} \tag{27}$$

The full implementation of the propulsion system is included below:

```
Im = 100000;
Tm = 10;
h1 = 0.1;       %same as main
```

```
persistent Qm  %set inital value for qm and keep it persistent
if isempty(Qm)
    Qm = 0;
end

Qd = rho * Dia^5    * KQ * abs(n_c) * n_c;
Qm_dot = 1/Tm * (Qd-Qm);
Qm = euler2(Qm_dot, Qm, h1);

n_dot = 1/(Im)*(Qm-Q);

xdot = [nu_dot' eta_dot' delta_dot n_dot]';
y = tau_total;
end
```

Problem 3d

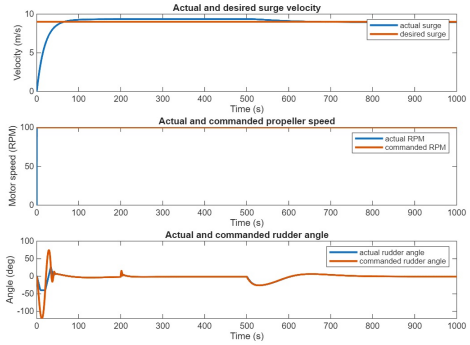
Under the assumption that the ship is cruising at a constant speed with zero rudder angle, we get $\dot{u} = \delta = 0$. Substituing this into Fossen[1] equation 9.33, we get:

$$(m - X_{\dot{u}})X_u - X_u = -X_{\delta\delta}\delta^2 + (1-t)T$$

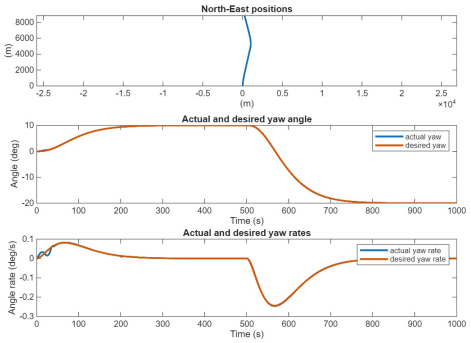
$$U = \frac{(t-1)T}{X_U}$$

Problem 3e

The system response after introducing open loop control can be seen in figure 6.



(a) Surge velocity, propeller speed and rudder angle after introducing open loop control



(b) North-east position, yaw angle and yaw rate after introducing open loop control

Figure 6: Simulation result for task 3e

This task utilized open loop control, which means that the system does not receive any feedback from the actual system during the control period. Therefore, any disturbance that are not modeled will not be accounted for and the system will not be able to adapt to any unforeseen changes happening during the control period. The controller used in this task does for example not take wind, waves or current into account. Hence, we will likely not achieve the desired speed if the heading angle is constant.

References

[1] T. Fossen, *Handbook of Marine Craft Hydrodynamics and Motion Control*. John Wiley & Sons, 2011.

[2] T. I. Fossen, "Ttk4190 assignment 2, part 2," Assignment description, NTNU, 2026, accessed: 26 October 2026. [Online]. Available: https://canvas.ntnu.no/